

Biomedical Knowledge Graph Triplet Extraction from PubMed Sentences

Comparing a SpERT Baseline with a Generative T5 Extension

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Project Goal

- Biomedical papers contain useful information, but most of it is written as unstructured text
- The goal was to extract subject–relation–object triplets
 - Example triplet:
AD | abbreviation_for | Alzheimer's disease
- These triplets can be used to help build biomedical knowledge graphs
- I compared:
 - SpERT (discriminative baseline)
 - T5-small (generative extension)

Example sentence and triplet:

Sentence: “Alzheimer’s disease (AD) is...”

Extracted triplet: AD | abbreviation_for | Alzheimer’s disease





Datasets and Exploratory Analysis

- **ADKG** focuses on Alzheimer's disease and related disorders
- **MDKG** focuses on mental disorders
- **MDKG** was denser than ADKG
- **ADKG** was more sparse, with many no-relation sentences
- MDKG also had longer-distance relations, which likely made RE harder

Data set	Total Sentences	Avg Entities/Sentence	Avg Relations/Sentence	Zero-Relation Sentences
AD KG	8,031	~2.6	~0.7	~65%
MD KG	6,678	~4.3	~1.6	~43–47%



Preprocessing and Pipeline

Raw Datasets Contained:

- sentence text
- entity annotations
- relation annotations

For SpERT:

- converted character-level entity spans to token-level spans
- converted relations into model-ready JSON

For T5:

- converted each sentence into an input/output text format

Built a Full Pipeline:

1. Load raw data
2. Run EDA
3. Convert data
4. Train models
5. Evaluate results
6. Perform error analysis

Baseline Model: SpERT

- Used **SpERT** as the main discriminative baseline
- Trained **separate models** for ADKG and MDKG
- Final encoder: **bert-small**
- Trained for **3 epochs**
- Evaluated:
 - NER micro F1
 - Relations without NEC
 - Relations with NEC

Setting	Value
Model	SpERT
Encoder	bert-small
Epochs	3
Train batch size	4
Eval batch size	2
Learning rate	5e-5

SpERT Test Results

Dataset	NER F1	RE F1 (without NEC)	RE F1 (with NEC)
ADKG	56.19	34.43	28.86
MDKG	67.55	28.76	24.95

- NER performance was stronger than relation extraction
- ADKG had slightly better relation extraction than MDKG
- MDKG had better NER, but relation extraction was harder
- Likely because MDKG had:
 - denser sentences
 - more entity types
 - longer-distance relations



Extension: Generative Triplet Extraction with T5-Small

- Reframed the task as **text generation**
- Input format:
Extract biomedical triplets from this sentence: [sentence]
- Target format:
subject | relation | object ; subject | relation | object
- If no relation was present, target = **none**
- Trained on a **combined subset**:
 - 3,000 train
 - 500 dev
 - 500 test

Match Type	Precision	Recall	F1
Strict	24.22	16.73	19.79
Relaxed	32.55	22.48	26.60

No-relation accuracy: **77.27%**



Comparison and Error Analysis

SpERT Strengths:

- good on abbreviation relations
- more reliable structured extraction

SpERT Weaknesses:

- struggled with indirect wording
- missed some multi-relation sentences

T5 Strengths:

- some exact triplets
- handled many no-relation examples well

T5 Weaknesses:

- hallucinated plausible relations
- sometimes generated incomplete outputs
- sometimes used the wrong relation type

Model	Evaluation	F1
SpERT	ADKG RE with NEC	28.86
SpERT	MDKG RE with NEC	24.95
T5-small I	Strict matching	19.79
T5-small I	Relaxed matching	26.60



Conclusions

- Built a full biomedical triplet extraction pipeline
- Completed:
 - EDA
 - Preprocessing
 - SpERT baseline
 - T5 generative extension
 - evaluation and error analysis
- Main takeaways:
 - **SpERT was better for exact structured extraction**
 - **T5 was a useful but more error-prone alternative**
- Future improvements:
 - try **BioBERT** or **PubMedBERT**
 - train T5 on more data
 - improve relaxed matching/evaluation

**Biomedical triplet extraction
is feasible, but relation
extraction is still the hardest
part.**

